

# **Basic Principles of Medicine 2**

## **Module: Endocrine & Reproductive Systems**

**DLA**

**Parturition and Lactation**

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# Required reading

- Medical Physiology – Principles for Clinical Medicine. Rhoades & Bell, 6e. Pages – 810 – 828. Chapter 38

## Objectives:

SOM.MK.II.BPM2.1.ER.1.PHYS.1090 Define parturition. Discuss the interaction of fetal and mechanical factors and role of hormones at initiation and throughout parturition.

SOM.MK.II.BPM2.1.ER.1.PHYS.1091 Discuss some causes of delayed or prolonged labor

SOM.MK.II.BPM2.1.ER.1.PHYS.1092 Explain the role of hormones in mammary gland development during puberty, pregnancy, and lactation.

SOM.MK.II.BPM2.1.ER.1.PHYS.1093 Explain the basis for the inhibition of milk secretion during pregnancy and the initiation of lactation after parturition.

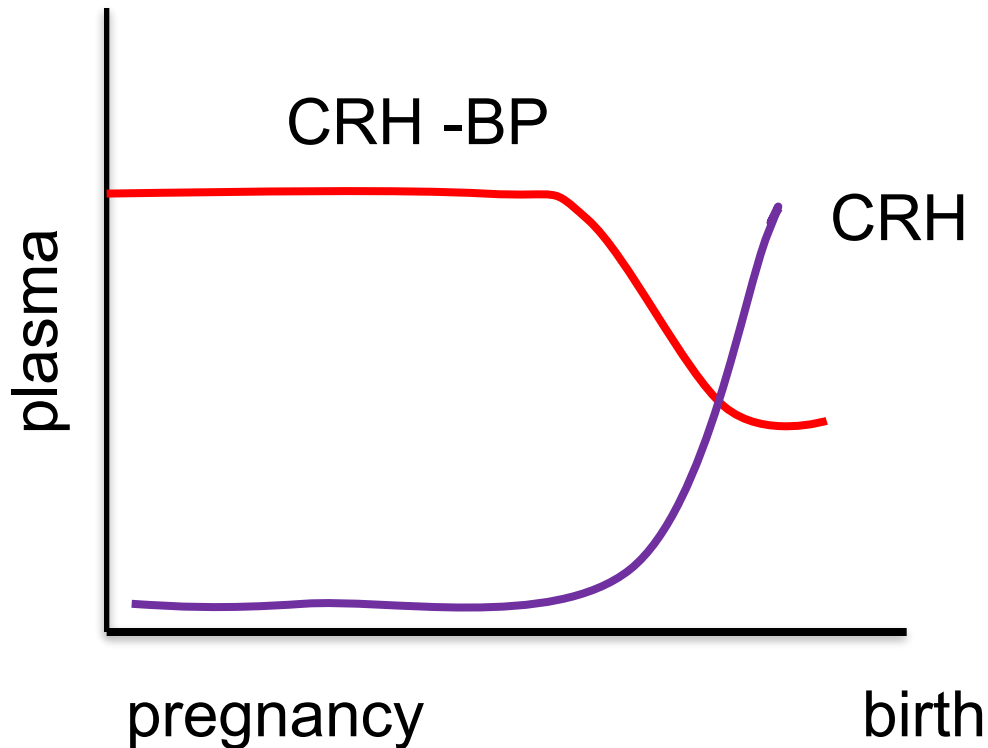
SOM.MK.II.BPM2.1.ER.1.PHYS.1094 Describe the neuroendocrine regulation of milk secretion and milk ejection & role of prolactin in contributing to anovulatory cycle

# Parturition

- Parturition means birth of the baby.
- The interactions of maternal & fetal factors regulate birth of the baby
- These include both mechanical & hormonal changes

# Changes before labor

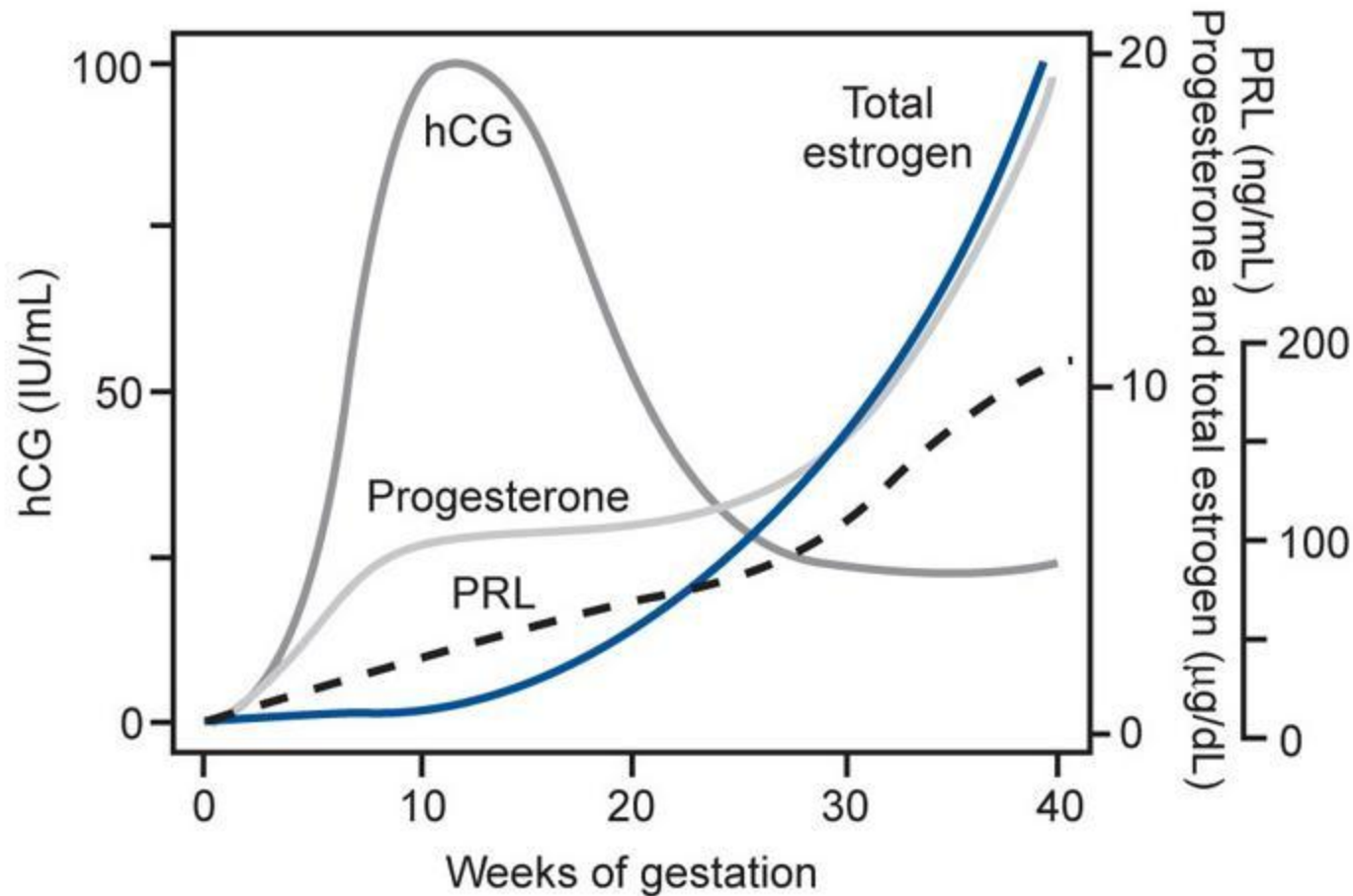
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- CRH secretion  
↑ exponentially  
near term
- At same time  
CRH- BP ↓ causing  
a steep rise in CRH

# The role of Placental CRH

- In fetus the CRH targets not only the pituitary gland but fetal zone of the adrenal gland to preferentially secrete DHEAS over cortisol
- Free CRH in maternal circulation increases maternal ACTH which in turn stimulates maternal synthesis of cortisol and androgens. These maternal androgens are converted to estrogen by the placenta.
- This results in  $\uparrow$  estrogen to progesterone ratio prior to parturition



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There is  $\uparrow$  in estrogen to progesterone ratio just prior to birth

# Factors– that increase contractility

- Uterus is quiescent during pregnancy as progesterone inhibits smooth muscle contraction thereby inhibits fetal expulsion
- Estrogens
  - ↑s the no of gap junctions in the myometrium
  - ↑s pg prostaglandin receptors in the myometrium
  - ↑s no. of oxytocin receptors in myometrium
  - estrogen ↑s uterine excitability
- Withdrawal of the progesterone block and elevated estrogen levels towards the end of gestation initiates parturition.

pg - prostaglandin

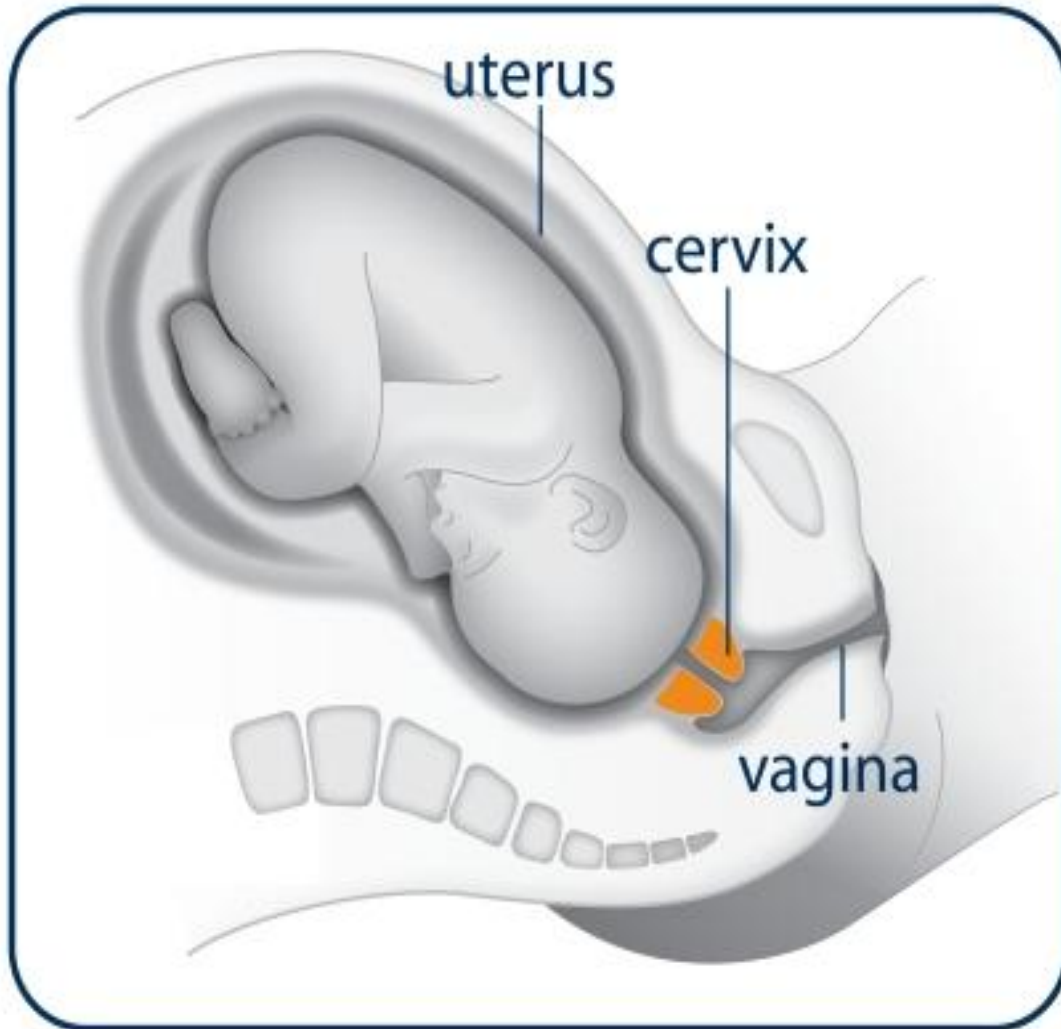


# Mechanical Changes

- Stretch of uterine smooth muscle leads to contraction
  - may stimulate myogenic reflex
- Stretch of the cervix
  - May stimulate neurogenic reflex which stimulates release of oxytocin

# Mechanical changes- myogenic reflex

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- intermittent fetal movements stretch the uterine muscle causing smooth muscle contraction
- twin pregnancy results in early labor

# Mechanical factors during labor

- There is coordinated contraction of uterus
- Just prior to labor uterus develops into functionally 2 distinct segments
- The upper segment is thicker & more muscular than the lower region
- Contractions occur from upper to lower segment
- Placenta is usually in the upper segment
- This facilitates the delivery of fetus 1<sup>st</sup>

# Mechanical factors during labor

- Lower segment is relatively thin & muscular & fuses with vagina
- Unification is observed as dilation of cervical external os from 1-2cm to 10cm
- Placenta previa is implantation of placenta in the lower segment
- Results in delivery of placenta before fetus & is life threatening for fetus as it comprises blood flow

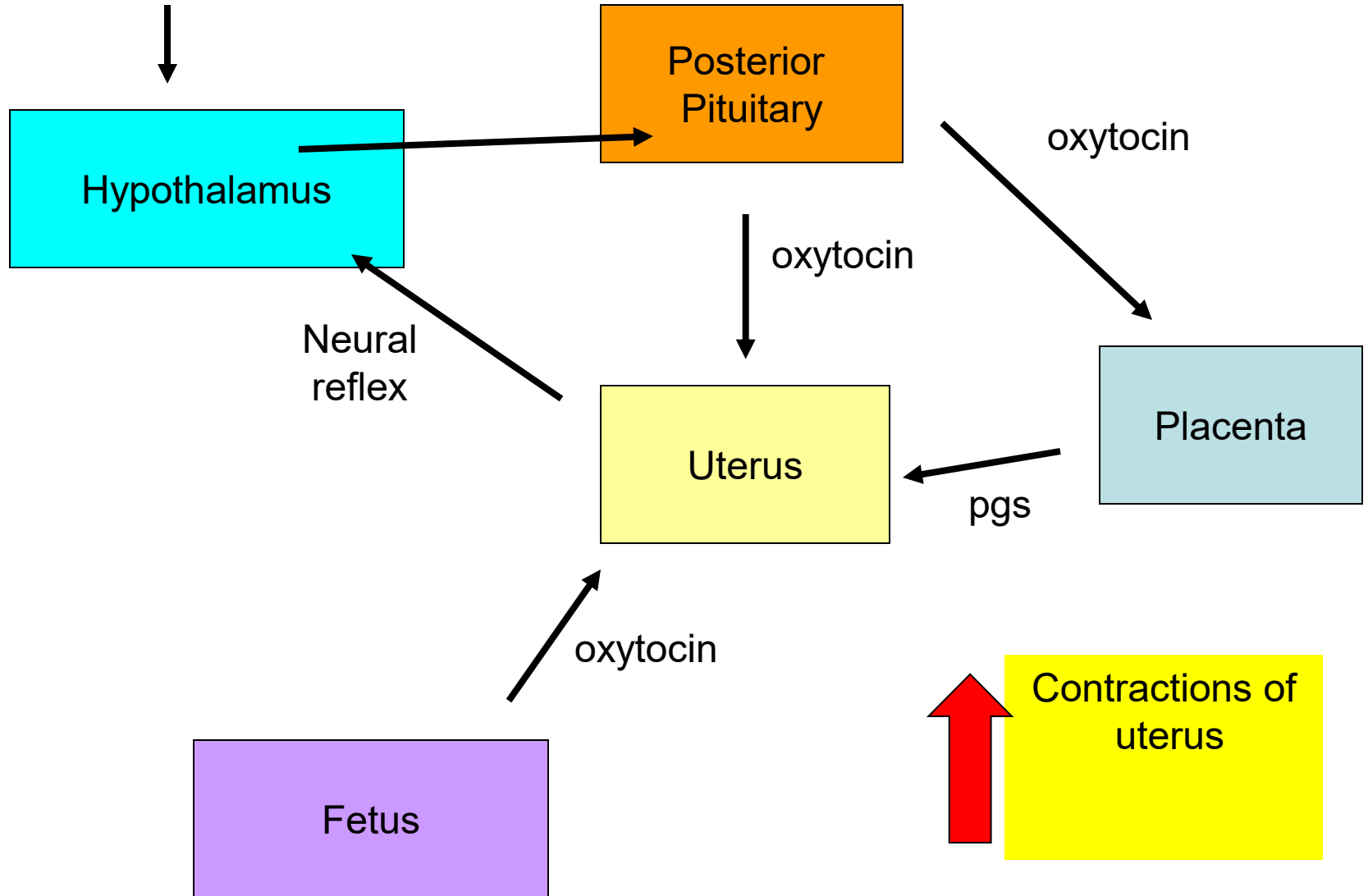
So what initiates labor???

# Estrogen progesterone ratio

- The **estrogen:progesterone** ratio increases closer to birth
- There is a rise in prostaglandin secretion
- $\text{Pg F}_2$  &  $\text{E}_2$  are potent stimulators of contractility, & cause significant ripening & dilation of cervix

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# Oxytocin in labor



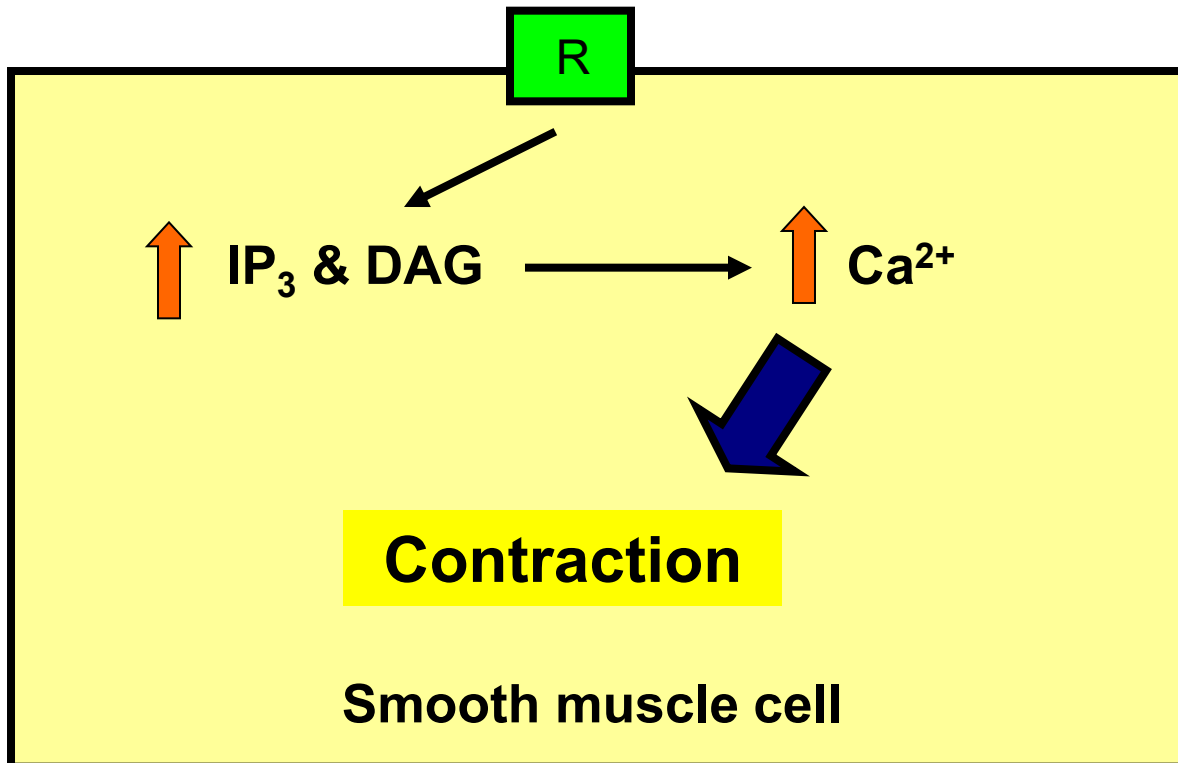
# Oxytocin in labor

- Oxytocin
  - ↑ secretion of oxytocin at labor
  - Stretch of cervix ↑ oxytocin release
- The fetus
  - Releases oxytocin, cortisol and pg' s
- ↑ oxytocin ↑ the intensity of uterine contraction



# Mechanism of action of oxytocin in uterus

## Oxytocin



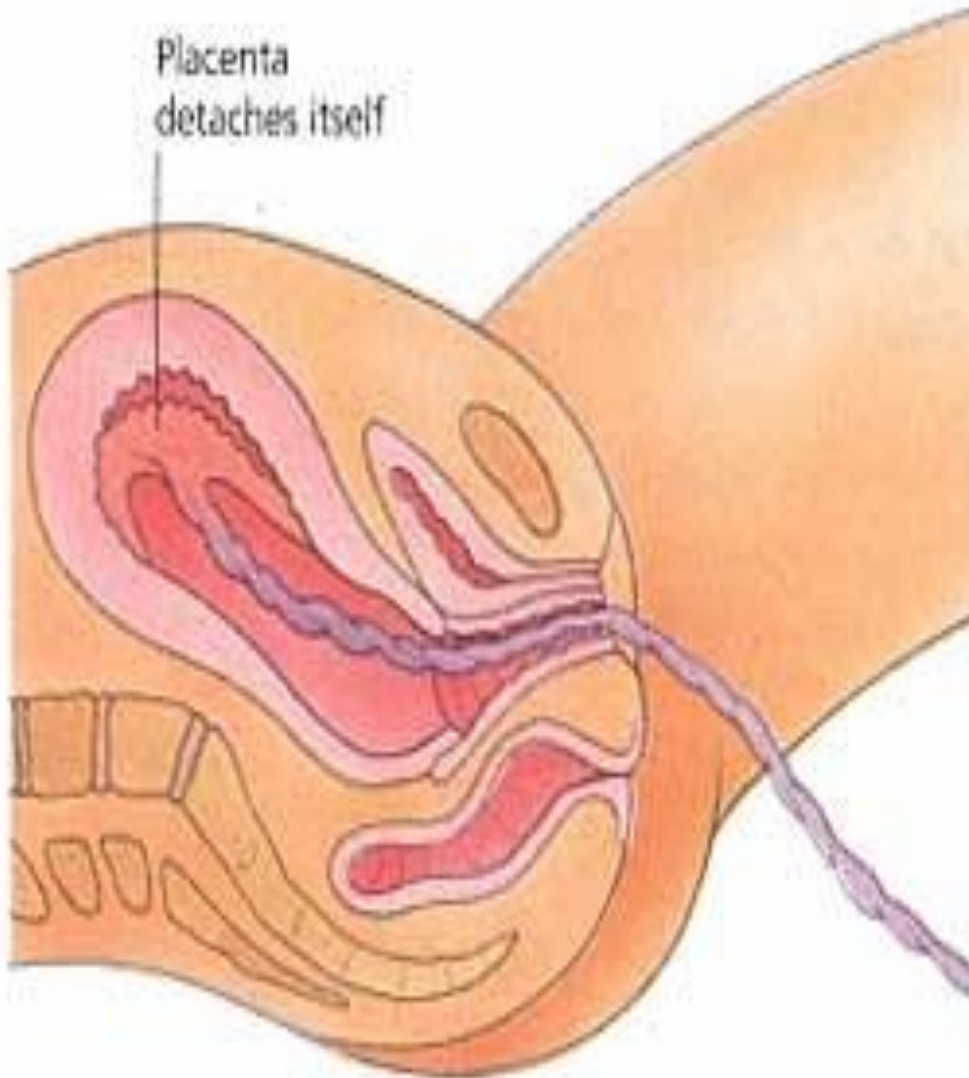
# Oxytocin in labor

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- ↑'s both amplitude & frequency of contraction
- Frequency of contractions is 2- 4 strong contractions every 10 minutes at the onset of labor
- Rhythmic contractions allows fetal blood to be oxygenated adequately
- Synthetic oxytocin (Pitocin) is sometimes used to augment or induce labor

# Placenta delivery

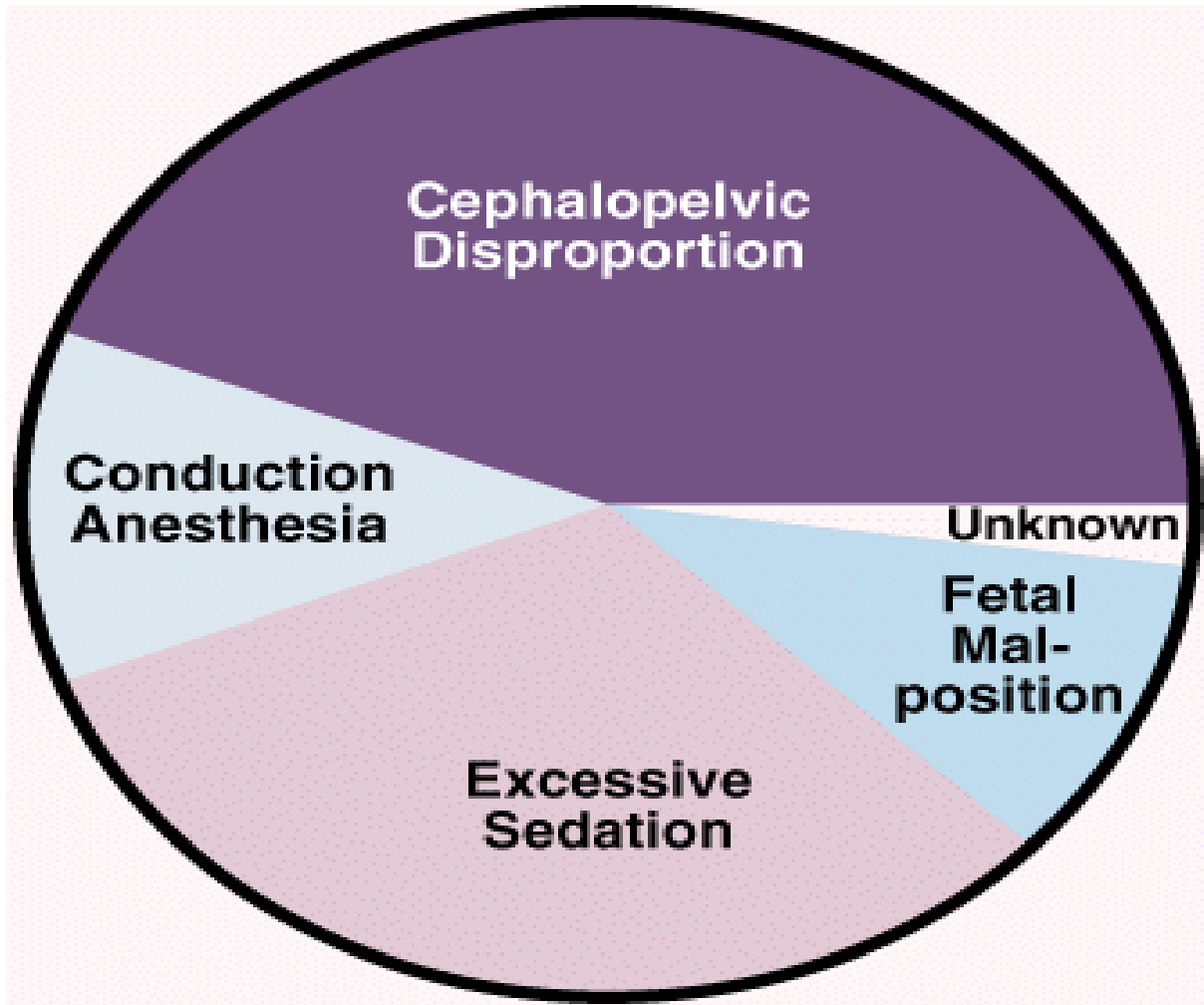
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Delivered 10 – 45 minutes later

- Uterus shrinks after the delivery of the baby
- This causes shearing effect between the walls of uterus & placenta separating it
- this causes bleeding but is limited by continued uterine contractions which cause vasoconstriction & vasospasm

# Causes of slow or delayed labor



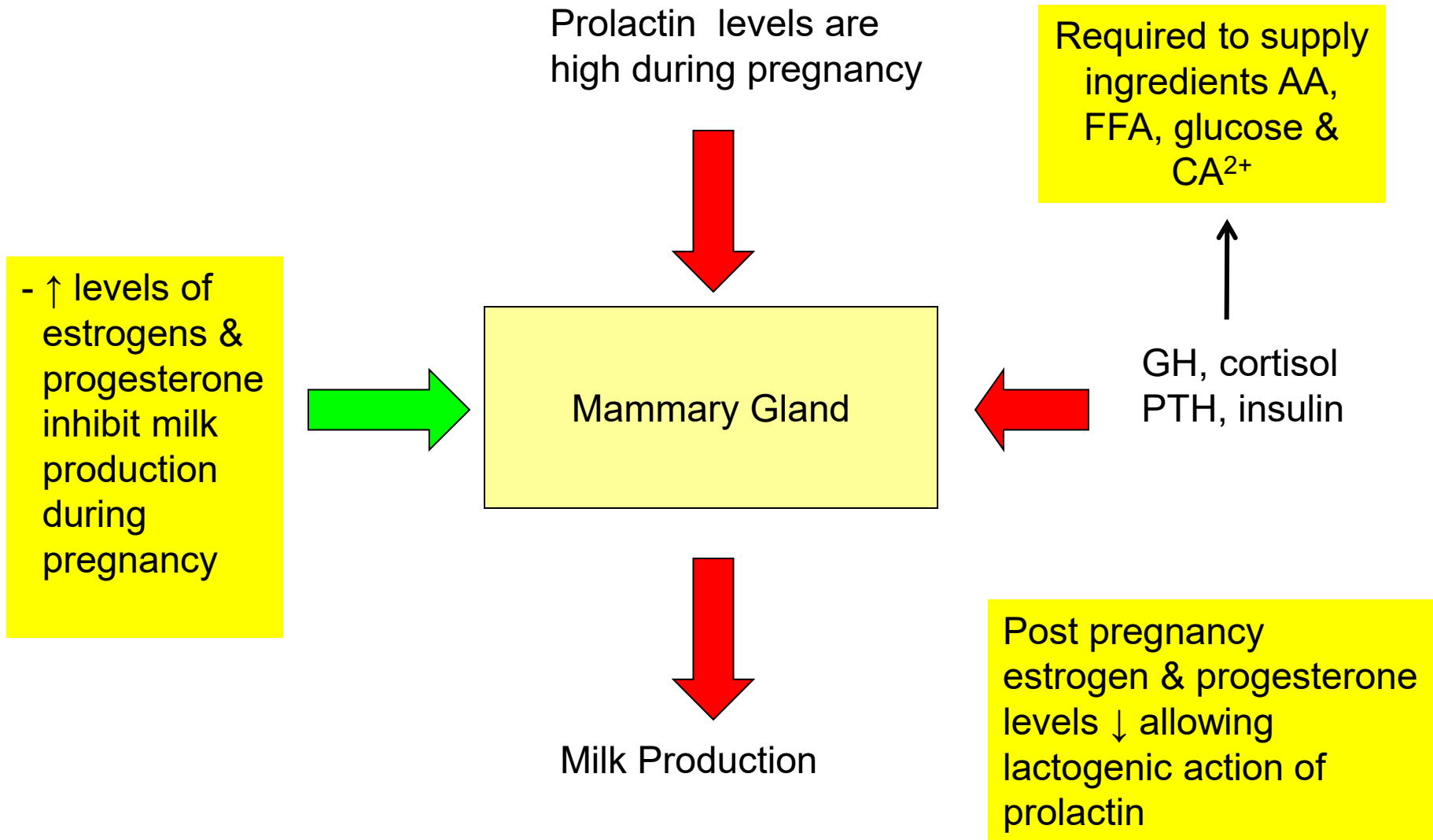
Lactation

# Breast Development

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- During Pregnancy
  - GH, PRL, cortisol and insulin help develop the ductal system
  - Estrogen is the major promoter
- Progesterone required to help with final growth of alveoli & its ability to secrete milk

# Role of hormones on the mammary gland



# Mechanism of prolactin release during lactation

**Producing  
more milk**

Suckling

Via mechanoreceptors in nipple

**Hypothalamus**



PRF



Dopamine  
inhibition

**Anterior  
Pituitary**



Prolactin

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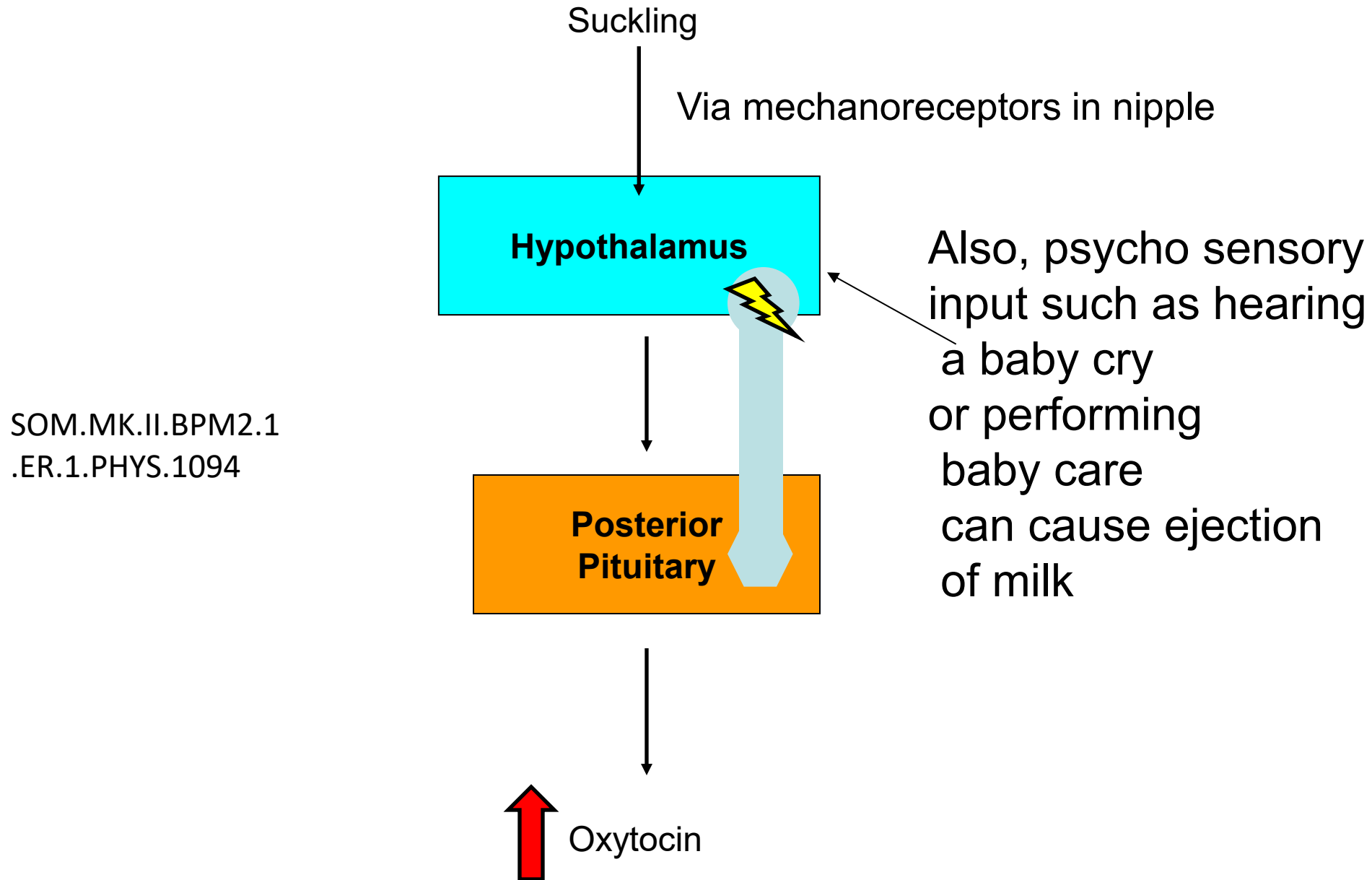
# Role of prolactin during lactation

- Prolactin is essential for lactogenesis (production of milk) and (galactopoiesis) maintenance of milk production.
- Initially  $\uparrow$  levels of prolactin will suppress the GnRH-pituitary-ovarian axis in the mother causing anovulatory cycles in females breast feeding the babies .
- However, this effect is transient after a few months the ovarian cycle is restored even though nursing of baby continues

# Milk ejection

- Milk is continuously produced by the alveoli but does not flow into the ductal system
- It must be ejected into the ducts before the baby can obtain it
- This is achieved by neurogenic mechanisms mediated through the hormone, Oxytocin. Oxytocin contracts myoepithelial cells of the alveoli to force milk into the ducts (milk ejection).

# Neurogenic mechanisms of Oxytocin release



# Milk secretion from mammary gland

- Milk secretion begins by the second postpartum day.
- Initially colostrum is secreted which persists for 3-5 days, with gradual conversion to mature milk.
- Colostrum is rich in proteins and minerals, and poor in fat and carbohydrates.
- Over next 2 to 10 days breasts secrete copious quantities of milk
- Besides providing nutrients to the baby it contains antibodies that serves protective function

# References

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